

CiteView: In-Situ Multi-Modal Citation Analysis for Scholarly Reading

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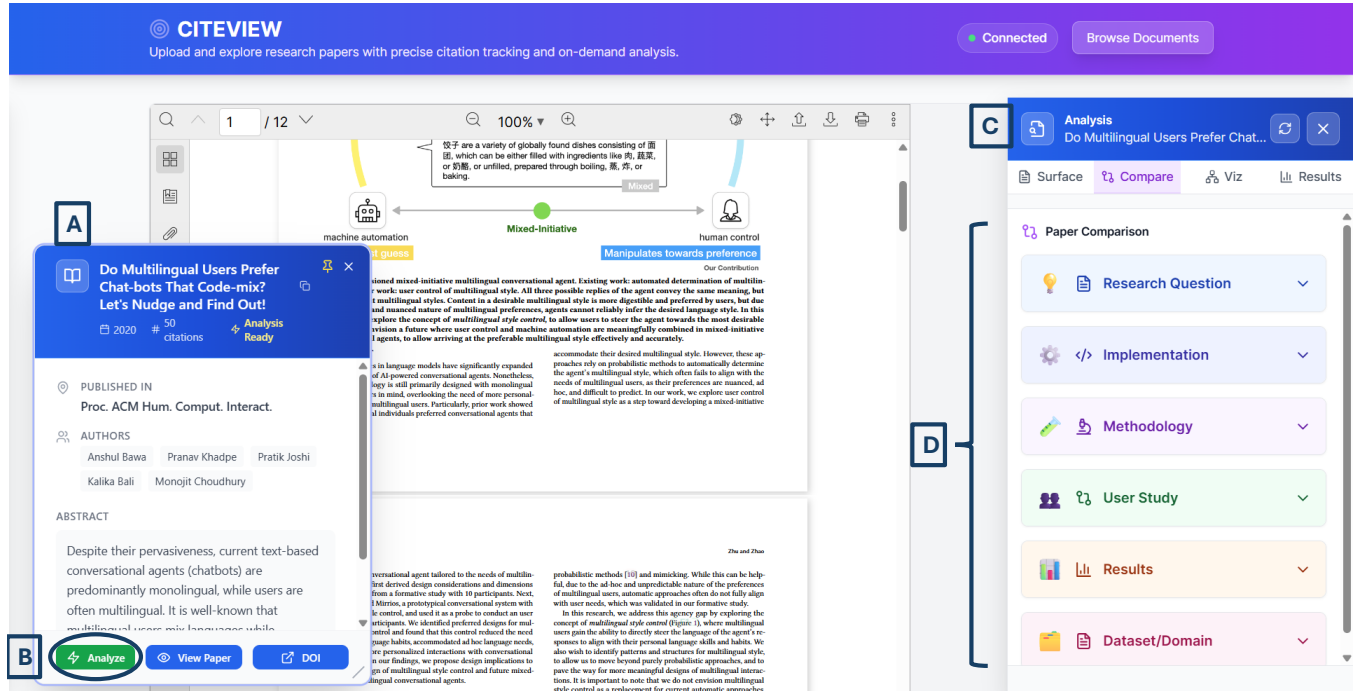


Figure 1: CiteView interface overview. Users upload a source paper; all inline citations become clickable. (A) Clicking a citation shows the cited paper’s metadata, retrieved automatically from scholarly APIs (no upload needed). (B) Analyze triggers, full-text retrieval, and multi-modal analysis. (C) Four-tab sidebar: Surface, Compare, Viz, Results. (D) Compare tab with six side-by-side dimensions. Note that throughout this paper, *source paper* is the paper being read; *cited paper* is a paper it references.

Abstract

Academic papers rely extensively on citations, yet understanding cited works often requires readers to leave their current context, imposing significant cognitive overhead. We present CiteView, an AI-assisted system that supports in-context exploration of cited papers within the PDF reading interface. CiteView provides four complementary views: *Surface* explains why a paper is cited and how it is used across the literature; *Compare* enables structured side-by-side analysis across six research dimensions; *Viz* presents semantic graphs and key figures; and *Results* extracts quantitative and

qualitative findings. The system employs a retrieval-augmented generation pipeline combining deterministic parsing, multi-component relevance scoring, and multi-agent LLM synthesis to ground all outputs in retrieved evidence. A user study ($n=12$) showed 97.9% task accuracy, positive usability (SUS: $M=71.9$), and low cognitive load (NASA-TLX Mental Demand: $M=2.0/10$). Participants preferred insight-driven views over descriptive summaries.

CCS Concepts

• Human-centered computing → Human computer interaction (HCI); Interactive systems and tools; User interface management systems.

ACM Reference Format:

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1 Introduction

Citations serve as the foundation of scholarly knowledge, linking new contributions to prior work and enabling readers to trace the

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evolution of ideas across research papers [12, 24]. However, verifying how a citation supports a specific claim requires leaving the current reading context: locating the cited paper, navigating access barriers, searching for the relevant content, and then returning to the source paper. As modern research articles routinely include dozens of references, this verification process becomes time-consuming and cognitively demanding, leading readers to either skip verification or divert substantial effort away from deeper analysis [1, 13].

Despite the importance of citations for scholarly sensemaking, existing tools provide limited support for accessing citation-specific context during reading [8, 17]. Text-based assistants such as SciSpace Copilot [25] clarify passages but cannot reveal how cited papers support specific claims, while citation network tools like VOSviewer [30] visualize bibliographic relationships without surfacing relevant content, figures, or evidence from cited papers themselves. As a result, readers must navigate between disconnected systems to interpret citations, imposing repeated context-switching costs. These limitations suggest a need for integrated citation analysis that surfaces contribution-level evidence directly within the reading interface.

To address this gap, we introduce CiteView (Figure 1), a system that integrates multi-modal citation analysis directly within PDF reading. While existing tools such as Semantic Reader [18], CiteSee [6], and CiteRead [23] focus primarily on textual summaries or personalized relevance signals, CiteView reveals contribution-level evidence through four complementary views grounded in distinct sensemaking needs [22, 26]: (1) *Surface*, providing context-aware explanations of citation intent to support orientation; (2) *Compare*, enabling structured comparisons across six research dimensions to support gap and overlap identification; (3) *Viz*, displaying extracted figures with semantic graphs to support mental model construction; and (4) *Results*, highlighting quantitative and qualitative findings to support evidence verification.

CiteView supports on-demand analysis of user-uploaded PDFs through automatic document parsing. Upon uploading a source paper, the system parses the document and makes all inline citations clickable. When a reader clicks a citation, CiteView displays a metadata card for the *cited* paper (Figure 1A), automatically retrieved from scholarly APIs. Clicking *Analyze* (Figure 1B) triggers retrieval of the cited paper’s full text and opens a sidebar (Figure 1C) containing the four analytical views. The system employs a retrieval-augmented generation (RAG) pipeline that combines deterministic document parsing (GROBID, PyMuPDF), multi-component relevance scoring, and multi-agent LLM-based synthesis to ground all outputs in retrieved evidence rather than free-form generation.

To illustrate, consider a reader encountering the sentence: “Building on the evaluation framework of [15], we adapted...” The reader may want to understand what that framework entails, how it was applied, and what empirical outcomes it produced. With CiteView, clicking [15] and selecting *Analyze* reveals: *Surface* summarizes that [15] introduces a task-based evaluation protocol for reading tools, which the source paper adopts; *Compare* highlights differences in study design (e.g., [15] reports 8 participants using think-aloud protocols, whereas the source paper uses 12 participants with questionnaires); *Viz* surfaces [15]’s system architecture diagram

with structured annotations; and *Results* extracts reported performance metrics (e.g., task accuracy and workload measures such as NASA-TLX). All of this is available without opening [15]’s PDF.

To assess the efficacy of CiteView, we conducted a mixed-methods study with 12 participants who performed targeted citation verification tasks. Participants achieved 97.9% accuracy, reported low cognitive load (Mental Demand: $M=2.0/10$), and rated usability positively (SUS: $M=71.9$). Qualitatively, participants reported that CiteView substantially reduced the overhead of citation exploration, enabling them to quickly interpret citation intent and access supporting evidence without context-switching.

This paper makes three primary contributions:

- **CiteView**, a system that integrates multi-modal citation analysis directly within PDF reading, combining contextual explanation, cross-paper comparison, visual evidence inspection, and results exploration through a retrieval-augmented generation pipeline.
- **A four-level analysis framework** (Surface, Compare, Viz, Results) grounded in sensemaking theory [26] that surfaces relationships between cited papers and their citing contexts, enabling multi-dimensional citation understanding within the reading interface.
- **Empirical validation** from a user study ($n=12$) demonstrating that integrating contextual explanations, visual evidence, and structured comparison supports effective citation verification with minimal cognitive load, alongside design implications for scholarly reading tools.

2 Related Work

2.1 Interactive Reading and Citation Tools

Scholarly reading involves iterative navigation, annotation, and cross-referencing rather than linear consumption. Early systems like LiquidText [28] supported active reading through spatial reorganization. Recent tools augment PDFs: ScholarPhi [11] surfaces definitions, Paper Plain [2] adds summaries, and Scim [8] highlights key sentences. These tools meaningfully improve local comprehension but share a common assumption: the document being read is solely the unit of analysis. They do not support evaluating the external works a paper cites. Some systems integrate citation context directly into reading. CiteSee [6] augments citations with personalized relevance indicators and supports on-demand access to cited papers, including unpublished manuscripts. CiteRead [23] surfaces commentary from subsequent work. PaperWeaver [17] and Threddy [13] support thread-based exploration, while ComLittee [14] and Synergi [15] enable network-level curation and cross-paper synthesis. Citation network tools such as CiteSpace [7] and VOSviewer [30] map bibliometric relationships across corpora. Collectively, these systems treat citations as pointers to related work, rather than as claims requiring verification; none are designed around the question a reader poses at the moment of encountering a specific citation claim.

2.2 AI-Assisted Document Understanding

Recent work has explored two complementary directions in AI-assisted reading: document-level summarization and retrieval-augmented grounding. LLM-based assistants such as SciSpace [25], Elicit [5],

and scite [20] summarize papers and classify citation statements as supporting or contrasting, but they treat documents as unstructured text and do not extract visual elements such as figures and tables. The Semantic Reader [18] represents the state of the art in AI-augmented reading interfaces, integrating citation cards and annotation layers directly into the PDF viewer. These systems demonstrate real value, yet they operate on pre-indexed corpora and surface information at the document level rather than targeting the specific claim a reader is trying to verify. On the retrieval side, RAG-based document interfaces have shown the value of grounding LLM outputs in retrieved evidence rather than parametric knowledge alone [16]. Recent work has also explored in-situ citation support by retrieving contextually relevant paragraphs from cited papers via similarity scoring. CiteView builds on this direction but treats the citation sentence as a structured query driving coordinated comparative, visual, and quantitative analyses directly tied to the citing claim. The distinction is not whether retrieval is used, but whether it is driven by document similarity or by the specific claim a citation is used to support. Prior work on annotation, discovery, and summarization leaves this gap unaddressed; CiteView closes it by organizing all analysis around the citation sentence itself, surfacing intent, comparison, visual evidence, and findings at the moment of encounter.

3 System Design

CiteView (Figure 1) transforms inline citations into interactive entry points for multi-modal analysis via four specialized views.

3.1 Design Rationale

CiteView’s design is inspired by empirical findings about how readers interact with citations during scholarly reading. Prior work has established that citation comprehension is not a simple lookup task; it involves assessing rhetorical purpose (why was this cited?), evaluating evidential support (what results back this claim?), and situating the cited work relative to the current paper (how do these approaches differ?) [1, 29]. Studies of scholarly sensemaking show that readers construct understanding through iterative cycles of foraging for relevant information, extracting key findings, and building schemas that organize relationships between sources [22, 26]. When these cycles require leaving the current document (to search for a cited paper, navigate access barriers, scan for relevant sections), the cognitive cost of context-switching disrupts the sensemaking process and degrades comprehension [13]. CiteView’s design directly targets these empirically documented pain points: the four-tab structure maps to the specific types of information that readers seek during citation comprehension, while the in-situ presentation minimizes the context-switching that prior work has identified as a key barrier [13, 23].

The four-tab framework is grounded in information foraging [21] and sensemaking theory [26]. Information foraging posits that users follow “information scent” to locate relevant content; sensemaking models describe how analysts build mental representations through cycles of search, extraction, and schema construction. We identified four distinct information needs when encountering citations, each formalized as a *Design Goal (DG)*:

- (1) **DG1 — Orientation:** “Why is this cited here?” Readers need the rhetorical role of a citation, corresponding to scent-following [21]. This motivates *Surface*.
- (2) **DG2 — Comparison:** “How does it relate to this paper?” Readers need to identify similarities and differences, corresponding to schema-based gap identification [26]. This motivates *Compare*.
- (3) **DG3 — Structural Understanding:** “What is its conceptual architecture?” Readers need to grasp the contribution structure without reading end-to-end, a need grounded in multimedia learning theory [19], which demonstrates that visual representations support comprehension that text alone cannot. This motivates *Viz*.
- (4) **DG4 — Evidence Verification:** “What results does it report?” Readers need direct access to empirical evidence, corresponding to information foraging toward high-value scent [21]. This motivates *Results*.

The structure emerged from three iterative rounds with three graduate student pilot testers. In the first round, we implemented six views. Pilot feedback revealed that *Citation Context* (the surrounding text in which a reference appears) and *Citation Purpose* (the rhetorical role of the citation) views were redundant, and *Figures* (visual elements extracted from the cited paper) and *Semantic Graph* (relational structure between concepts) views were treated as complementary. In the second round, we consolidated to four views: *Surface* (merging context and purpose), *Compare*, *Viz* (merging figures and graph), and *Results*. In a third validation round, we confirmed all common citation questions could be answered using four tabs, while removing any single tab left gaps.

The Compare tab’s six dimensions (research question, methodology, implementation, user study, results, dataset/domain) reflect the IMRAD structure [27] standard in empirical HCI and CS papers (Introduction, Methods, Results, and Discussion). During pilot testing, more than ten dimensions were too granular, while 3 conflated distinct concerns. These dimensions are calibrated for empirical systems papers and may need adaptation for other traditions (see Section 8).

3.2 Surface Tab

The Surface tab (Figure 2) answers: “Why is this paper cited here, and what does it add?” It combines the citation context with ranked sentences from the cited paper to generate four evidence-grounded bullet points using GPT-4¹. Synthesized explanations are constrained to the retrieved evidence to reduce hallucination.

For example, given “Following [15]’s annotation framework, we designed our interface,” the bullets might state: (1) [15] introduced a three-layer annotation model for active reading on tablets; (2) Validated with 24 students showing 40% faster retrieval; (3) The current paper adapts the highlight layer with AI overlays; (4) [15] focused on single-document annotation; this work targets cross-document contexts. These connect specific elements of [15]’s contribution to the citing sentence’s claim, moving beyond a generic summary.

Bullets use progressive disclosure: compact by default, allowing users to expand and access the full underlying text and additional details on demand (Figure 2A). The tab also visualizes how the cited

¹OpenAI GPT-4: <https://platform.openai.com/docs/models/gpt-4>

work is typically used across the literature (Figure 2B), comparing the current citation’s purpose against usage patterns. For example, if a paper is typically cited for methodology (e.g., 60%) but the source paper cites it as Results Support, this divergence is immediately visible. Users can explore example citations from other papers, each expandable to show the original passage (Figure 2C).

3.3 Compare Tab

The Compare tab (Figure 3A) enables side-by-side comparison across six research dimensions. The rationale for these dimensions is described in Section 3.1. Comparison is valuable when evaluating whether cited work uses similar methods or produces complementary findings. For example, when a source paper cites [15] as methodological precedent, the Compare tab can reveal that [15] used a between-subjects design with 8 participants while the source paper uses a within-subjects design with 20. Such differences are often obscured in citation sentences. Comparisons appear as collapsible side-by-side cards (source left, cited right). When content is missing, the system indicates absence rather than generating speculative content.

Distinction from the Results tab. The Compare tab’s Results dimension provides a *comparative summary* (e.g., “both report improved accuracy, but [15] found higher load”), supporting quick and easy cross-paper comparison. In contrast, the Results tab focuses on *deep extraction* of specific metrics and evidence units, enabling a more detailed understanding of the paper’s findings. One is relational and lightweight; the other is detailed and analytic. Pilot testing confirmed distinct usage patterns.

3.4 Viz Tab

The Viz tab addresses the challenge that grasping a cited paper’s structure normally requires reading it end-to-end, a limitation explained by multimedia learning theory [19], which holds that visual representations support comprehension that text alone cannot achieve. It provides two visual abstractions: (1) a *Semantic Graph* (Figure 3C), an interactive node-link diagram with eight types (Problem, Approach, System, Finding, Dataset, Insight, Background, Critique) connected via rule-based heuristics and semantic similarity. For example, for a cited paper on “interactive reading tools,” the graph shows: Problem (“readers lose context”) → Approach (“in-situ augmentation”) → System (“sidebar with summaries”) → Finding (“reduced completion time by 30%”). Users click nodes to reveal detailed explanations. (2) a *Figure View* (Figure 3B) surfacing three curated figures (teaser, primary contribution, contextually relevant) with structured explanations (Purpose, Key Components, How It Works). All views operate on a single cited paper; multi-paper visualization is left for future work.

3.5 Results Tab

The Results tab (Figure 3D) automates extraction of findings into three views: *Quantitative* extracts metrics, values, baselines, and significance, e.g., “accuracy: 92% (vs. baseline 78%, $p < 0.01$).” *Qualitative* identifies themes and user behaviors as claims paired with evidence. *Relation* maps how cited results connect to the source paper through four types: validation, comparison, extension, or critique.

4 Implementation

CiteView is a full-stack web system (React/TypeScript frontend, Python/Flask backend).

Document Processing. PDFs are rendered via PDF.js². Citations are extracted using GROBID³ (bibliographic structure) and PyMuPDF⁴ (coordinate detection). This dual approach is necessary because GROBID provides structured citation metadata without positional information, while PyMuPDF provides precise text coordinates but lacks citation semantics. Metadata is enriched via the Semantic Scholar API. Full texts of cited papers are retrieved from open-access endpoints (Semantic Scholar⁵, Unpaywall⁶, arXiv⁷, PubMed Central⁸); if unavailable, the system degrades to metadata-only analysis and explicitly informs the reader that detailed positional or contextual information could not be retrieved.

Retrieval-Augmented Generation. When a citation is clicked, the surrounding sentence is captured as the query context. The cited paper is segmented into sentences, ranked using four signals: (1) semantic similarity (sentence-transformer embeddings, all-MiniLM-L6-v2); (2) keyword overlap (TF-IDF); (3) section importance, weighting Introduction, Methods, and Results higher to prioritize contribution-bearing content; and (4) position weighting favoring earlier sentences, reflecting academic writing conventions where key claims are front-loaded [27]. The top 8 sentences are passed to GPT-4 with structured prompts. Surface prompts generate 4 bullet points; citation purposes are classified into 6 types [29]; the Compare and Results tabs use structured JSON output validated against schemas.

Multi-Agent Analysis. The Compare and Results tabs employ a CrewAI⁹-based multi-agent system. Compare uses an Ingestion Agent for structured extraction and an Analyst Agent for generating paired descriptions. The Results tab uses three specialized agents for quantitative, qualitative, and relational analysis. Outputs are constrained through evidence-grounding, structured formats, external validation (Semantic Scholar labels), explicit refusal instructions, and rule-based fallbacks to prevent unsupported outputs.

Figures & Semantic Graphs. Figures are extracted via PyMuPDF and ranked by position and caption-to-context similarity. Semantic graphs use GPT-4 to generate section summaries mapped to a JSON schema (8 node types, 5–10 nodes). Edges use rule-based heuristics and cosine similarity (> 0.6). Rendered with D3.js¹⁰ force-directed layout.

Pipeline Quality. During development, we assessed CiteView’s outputs across 15 citation analyses covering 5 HCI papers. All outputs were manually evaluated by two researchers. Surface bullet points were assessed for factual grounding: all 60 bullet points (60/60) were judged as factually accurate and contextually specific; however, approximately 12% (7 bullet points), while factually correct, were overly generic.

²PDF.js: <https://mozilla.github.io/pdf.js/>

³GROBID: <https://github.com/kermitt2/grobid>

⁴PyMuPDF: <https://pymupdf.readthedocs.io/>

⁵Semantic Scholar: <https://www.semanticscholar.org/>

⁶Unpaywall: <https://unpaywall.org/>

⁷arXiv: <https://arxiv.org/>

⁸PubMed Central: <https://www.ncbi.nlm.nih.gov/pmc/>

⁹CrewAI: <https://www.crewai.com/>

¹⁰D3.js: <https://d3js.org/>

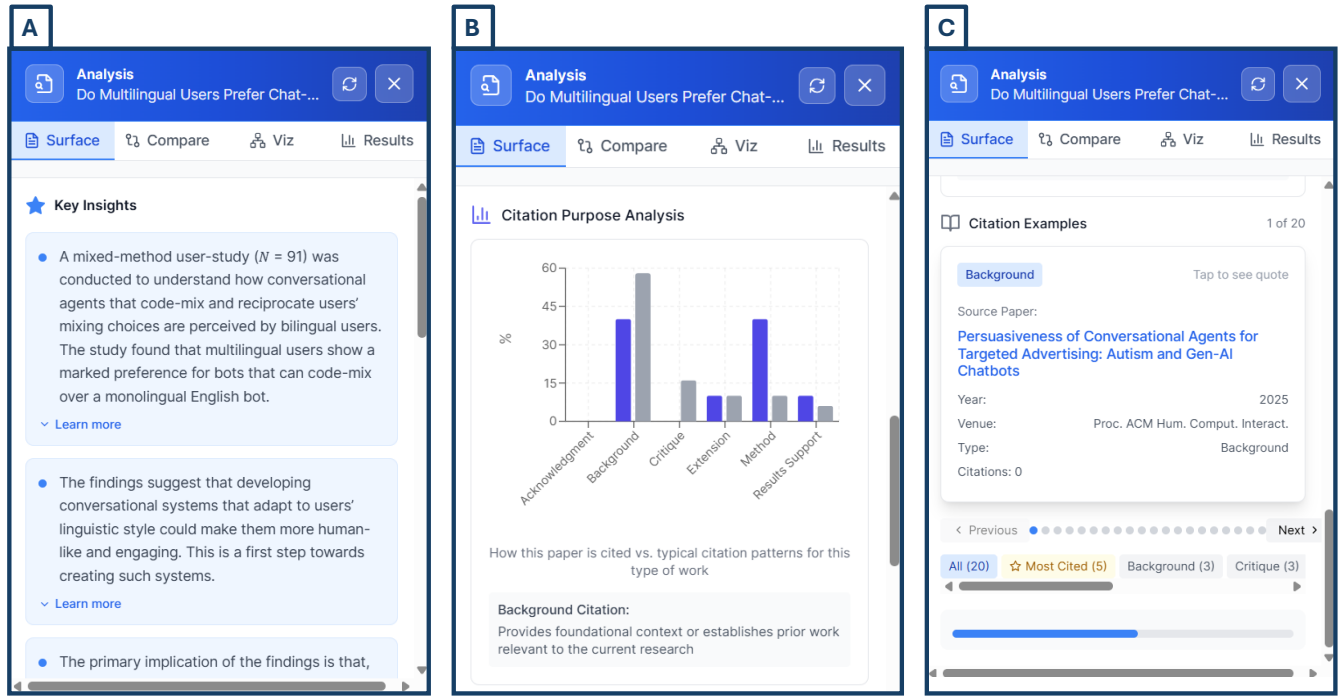


Figure 2: Surface tab. (A) Context-aware bullet points explaining the cited paper's role; (B) citation-purpose distributions comparing current use against typical patterns; (C) example citing papers with expandable quotations.

For the Results and Graph views, extracted information was similarly verified against the source papers and found to be consistently accurate across cases, with only a small number of cases (2–3 instances) producing more generic descriptions rather than detailed insights. The Compare tab dimensions were evaluated against manual reading. Across 90 dimension pairs (15 citations $\times$ 6 dimensions), none contained inaccurate information; however, approximately 20–25 cases (22–27%) produced more generic descriptions. These primarily occurred in the user study and results dimensions when the cited paper itself provided limited detail, and in the dataset dimension, where many HCI papers do not define a formal dataset. In such cases, the system correctly indicated absence rather than introducing unsupported content. At the citation level, all 15 analyses produced multiple accurate outputs, indicating that the pipeline reliably surfaces useful information for each cited work. These assessments informed iterative prompt refinement but do not constitute a formal evaluation; systematic pipeline validation remains future work (see Section 8).

5 User Study

We conducted a mixed-methods study ($n=12$) approved by our Institutional Review Board (IRB).

5.1 Participants

We recruited 12 participants (4 women, 8 men) aged 23–26 ($M=24.5$), including 10 master's and 2 senior undergraduates from technical disciplines (computer science, physics, management). All routinely read academic papers (self-reported frequency: master's students

reported reading 2–3 papers per week; undergraduate students reported approximately one paper per week from their second year onward). All participants were familiar with PDF reading tools such as Zotero and browser-based reference management extensions. Participants were recruited through university mailing lists and completed one-hour sessions in a controlled lab setting. Three were randomly selected to receive \$20 gift cards as compensation.

5.2 Study Design and Procedure

The session began with informed consent, followed by a guided introduction to CiteView's features. Participants then freely explored the system with a sample paper before proceeding to the main tasks.

Source papers. We selected three recent HCI papers as source materials, each containing 30–50 citations with diverse citation purposes. To ensure breadth, the papers spanned three distinct HCI domains: multilingual conversational agents, neuroadaptive learning interfaces, and chart-text linking in documents. Each paper was assigned to four participants in a between-subjects design; all participants completed tasks individually. While the papers broadly matched participants' technical backgrounds, none directly aligned with their specific subfields, ensuring that responses reflected genuine exploration rather than prior knowledge. For each paper, we pre-selected citations that were openly accessible and contained clear results and figures.

Pilot baseline condition. Our evaluation follows the single-system study methodology common in UIST and CHI systems

research, where the primary goal is to establish that a new interaction paradigm is usable, that its outputs are accurate, and that users can extract value from it. Prior systems introducing new reading and sensemaking interfaces, including Threddy [13], Synergi [15], LiquidText [28], and the Semantic Reader [18], adopted this approach, measuring task performance and usability within the system rather than comparing against traditional workflows in a controlled baseline condition. We adopt the same strategy: our study assesses whether CiteView surfaces correct information, whether participants can locate relevant evidence across the four views, and whether the interface imposes an acceptable cognitive load.

To calibrate expectations, we conducted a pilot with three graduate students who performed the same citation tasks using their usual tools (Google Scholar, institutional library access, standard PDF readers). Without CiteView, participants spent 10-15 minutes per citation: locating the cited paper, navigating access barriers (paywalls, institutional login redirects), finding relevant sections, and manually extracting information, totaling 20-30 minutes for two citations alone. The equivalent CiteView tasks took approximately 10-20 minutes, including two citation analyses. This gap confirmed that a within-session baseline would introduce fatigue confounds: the full baseline protocol would have required roughly 90 minutes, double the CiteView session length. Rather than compress the baseline into a superficial walkthrough, we opted for a more focused design, evaluating CiteView thoroughly and collecting retrospective comparisons through post-study interviews, where participants contrasted CiteView against their established workflows. Additionally, no directly comparable system exists that supports the same citation analysis workflow, making a tool-to-tool controlled comparison infeasible.

Task blocks. The study comprised three blocks of four multiple-choice questions each. In Blocks 1 and 2, participants were directed to specific tabs to enable systematic evaluation of each view's effectiveness: Block 1 assessed the Surface and Compare tabs through questions on citation context, purpose, and cross-paper relationships, while Block 2 assessed the Results and Viz tabs through questions on empirical findings and visual evidence. Both blocks used the same citation. Block 3 introduced a different citation with unrestricted tab access, assessing participants' ability to independently select appropriate views for each question type. Participants could rest between blocks and skip any question they were unable to answer.

Question design. Questions were generated by analyzing citations with CiteView, then formulating items verifiable from displayed content but not immediately obvious. Correct answers were verified by cross-referencing against original cited papers. Table 1 shows representative examples from each block.

Questions assessed two dimensions: (1) *content accuracy*, whether CiteView surfaces correct information, and (2) *navigation efficiency*, whether participants locate relevant content across the multi-view interface. For instance, "What does the main figure illustrate?" tests the figure extraction pipeline through AI-generated explanations, while "How is this paper usually cited?" tests CiteView's synthesis of citation-purpose distributions from external data. Blocks 1 and 2 isolated content accuracy per view by directing participants to

Table 1: Representative questions from each task block across the three source papers. Block 1–2 directed participants to specific tabs; Block 3 required independent tab selection. Each participant answered 12 multiple-choice questions (4 per block); stems are shown here without answer options.

#	Example Question	Target Tab
Block 1: Tab directed (Surface & Compare)		
1	Why was this cited paper included at this point in the source paper?	Surface
2	How is this paper usually cited in other research?	Surface
3	How do the user studies differ between the source and cited papers?	Compare
4	How do the research questions of the source paper and the cited paper differ?	Compare
Block 2: Tab directed (Results & Viz)		
1	According to the quantitative findings, how did the system perform compared to the baseline?	Results
2	What themes did participants report in the qualitative findings?	Results
3	According to the extracted figure, how does the system connect text and tables?	Viz
4	What does the extracted figure illustrate about learning performance across groups?	Viz
Block 3: Unrestricted (Any Tab)		
1	Why is this cited paper relevant to the source paper's contribution?	Any
2	How does the cited paper's methodology fundamentally differ from the source paper's?	Any
3	What gap in prior work does the cited paper identify?	Any
4	How did the system's performance compare across conditions?	Any

specific tabs; Block 3 assessed both dimensions under naturalistic, unrestricted navigation. This design evaluates usability and accuracy, not reading comprehension or sensemaking depth (see Section 7). Following the task blocks, participants completed the NASA-TLX [10] to assess cognitive load and the SUS [4] to measure usability. We also included three supplementary items, Time Efficiency, Ease of Finding Information, and Overall Helpfulness, following Hart's guidance on domain-specific adaptation of workload measures [9]; these are reported separately from the standard NASA-TLX subscales. Each session concluded with a 20-minute semi-structured interview. Table 2 presents the semi-structured interview questions administered after all three task blocks.

5.3 Measures

We measured task performance (percent correct across all blocks), interaction logs (time per tab, switching frequency), perceived workload (NASA-TLX [10]), and usability (SUS [4]). Semi-structured interviews were analyzed via inductive thematic analysis [3].

6 Results

Results are reported separately for tab-directed blocks (1–2) and the unrestricted block (3), reflecting their distinct assessment goals: per-view accuracy versus naturalistic multi-view navigation.

6.1 Task Performance

Overall accuracy was 97.9% (141/144 correct; $M=11.75/12$, $SD=0.45$), with nine participants achieving a perfect score (12/12).

Guided blocks (1–2): Performance was high in the guided setting, with a combined accuracy of 97.9% (94/96). Block 1 achieved

Table 2: Semi-structured interview protocol administered after all task blocks. Questions are organized by assessment focus; each targets a distinct construct used in the thematic analysis (Section 6.3).

#	Question	Construct
System Experience & Learnability		
1	Did the tool meet your expectations? Why or why not?	Expectations
2	How easy or difficult was it to find information across the four views?	Navigation
3	Were there any features you found confusing? If so, which ones and why?	Learnability
4	How long did it take you to feel comfortable using CiteView?	Learning curve
Content Quality & Trust		
5	How accurate was the information provided? Did it display what you expected? If not, what was missing?	Accuracy
6	How confident were you that CiteView's information was correct?	Trust
7	Were the Viz tab's visual elements (figures, semantic graphs) helpful or distracting?	Visual utility
Workflow Integration & Adoption		
8	Which of the four views (Surface, Compare, Viz, Results) did you find most/least useful? Why?	View preference
9	How did the four views work together when answering questions? Did you need multiple views?	Complementarity
10	How did CiteView differ from your usual workflow for looking up cited papers?	Comparison
11	Do you see this tool fitting into your academic reading workflow? Why or why not?	Adoption
12	What would you change or add to make CiteView more useful?	Improvement

95.8% (46/48), while Block 2 reached perfect accuracy (48/48). Errors in these blocks were rare and primarily occurred in comparison-oriented questions (e.g., distinguishing research questions or user study setups), rather than simple information retrieval.

Unrestricted block (3): maintained similarly high performance at 97.9% accuracy (47/48), with only one error. This indicates that participants could effectively navigate and utilize the multi-view interface independently, without explicit guidance on tab selection. Interaction logs showed strategic tab selection by question type, with no single view dominating usage.

Tab usage in Block 3. When participants freely chose which tabs to consult, their selections aligned with question type: questions about empirical findings (e.g., "What metrics were reported?") predominantly led to the Results tab; questions about methodological differences led to Compare; questions about paper structure led to Viz; and questions about citation purpose led to Surface.

Critically, all tabs were actively used across sessions, no tab was unused, and no single tab dominated the interaction. Across two citations, participants spent approximately 10–20 minutes in total, distributing their time across views based on question type. Results tab received the most engagement (~4–5 minutes, ~35–40% of session time), followed by graph-related views (~5 minutes, ~35%), Compare (~3–4 minutes, ~25%), and Surface most briefly (~2 minutes, ~10–15%), suggesting quick orientation use rather than deep reading. This distribution provides behavioral evidence that each view serves a distinct role, supporting quick orientation, structured comparison, and deeper exploration of results and figures, reinforcing the complementary design rather than functional

redundancy. Additionally, several sessions showed repeated switching between tabs, suggesting that some citation questions benefit from triangulation across multiple views.

6.2 Usability and Workload

SUS: $M=71.9$ ($SD=13.2$), above the 68 benchmark [4]. Positively worded items received strong ratings (intention to use: $M=4.25$; ease of use: $M=4.33$; feature integration: $M=4.50$; learnability: $M=4.17$; confidence: $M=4.33$). Negatively worded items were low (complexity: $M=2.08$; need for support: $M=2.25$; inconsistency: $M=1.67$; difficulty: $M=1.83$) (Figure 4).

NASA-TLX: On the NASA-TLX (0–10 scale), participants reported low Mental Demand ($M=2.0$, $SD=1.0$), Effort ($M=2.1$, $SD=0.8$), and Frustration ($M=2.2$, $SD=2.4$), suggesting high individual variability. Most participants reported minimal frustration, while a small number found specific features confusing. Supplementary items indicated strong perceived support: Time Efficiency ($M=8.9$), Ease of Finding Information ($M=9.1$), and Overall Helpfulness ($M=9.2$) (Figure 4).

6.3 Qualitative Analysis

Four themes emerged from thematic analysis [3]:

Theme 1: Reduced context-switching overhead. Participants consistently identified specific steps in their traditional workflows that CiteView eliminated. P3 described the conventional process: "go on Google, figure out if I had access to that paper... log into my student account, read the paper... and then confirm that it's the paper that's been cited in the references." With CiteView: "I just have to click the citation." P1 noted that CiteView provided contributions "in short form... the main contributions," making them available "instantaneously" rather than requiring manual extraction from lengthy documents. P4 estimated the efficiency gain as "reduced it by 10X." P2 stated that with CiteView, "I can read more papers within a short period of time," suggesting CiteView enabled lightweight verification that participants would otherwise skip entirely due to time constraints.

Theme 2: Preference for insight-driven views. Compare, Results, and Viz were consistently preferred over descriptive summaries, and participants articulated *specific* reasons for each preference. P1 chose Results as their favorite "because I can instantly understand what the cited paper did," specifically valuing the structured extraction of metrics and findings rather than needing to scan dense results sections. P2 highlighted Compare because the side-by-side structure made it possible to "see where the two papers differ" on methodology and study design, which P2 noted would otherwise require opening both PDFs simultaneously. P3 emphasized the Viz tab "because it provides insights really quickly... and highlighted the important graphs or figures," enabling rapid assessment of whether a cited paper's visual evidence was relevant. P6 called the citation-purpose visualization "a very smart decision" because it reveals "what this cited paper was used for the most, or cited in other papers for," providing "good relevant data that people can use within their actual paper." However, P4 suggested that the Surface view could be made even more concise, expressing a preference for very short summaries, such as keywords or highlighted points, rather than longer text.

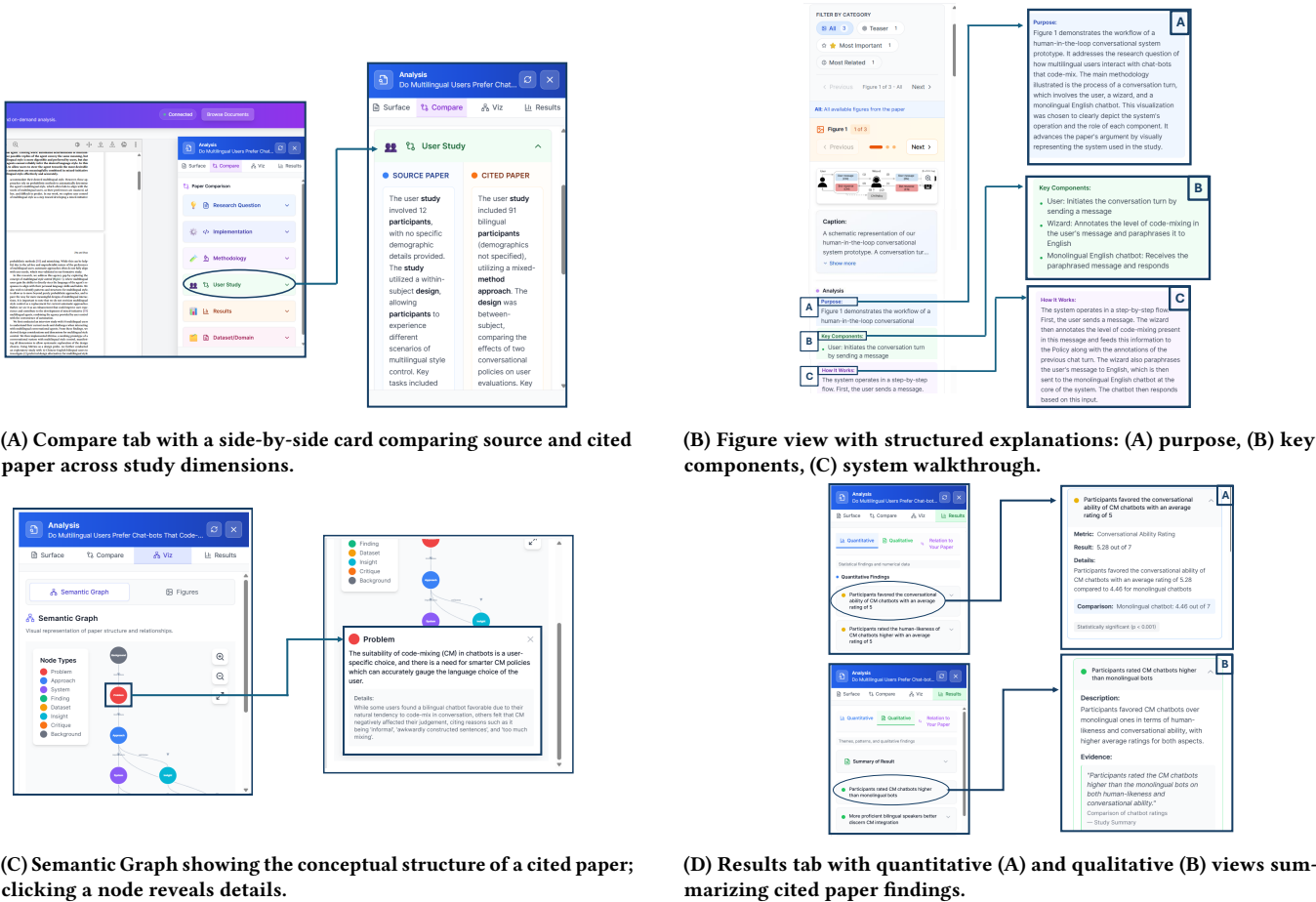


Figure 3: CiteView’s four information views: (A) Compare, (B) Figure, (C) Semantic Graph, and (D) Results, each supporting a distinct reading strategy without leaving the reading context.

Theme 3: Feature Richness and Progressive Disclosure. Participants provided constructive feedback regarding information density and feature organization. While the richness of features was broadly appreciated, some participants noted that the number of available views could initially feel overwhelming. For example, P5 described the system as “amazingly designed” but noted it was possible to “get lost within the features.” At the same time, participants strongly valued the breadth of functionality: five participants (P6–P10) described all features as “crucial and relevant,” with P6 emphasizing they would “add more tools rather than getting rid of any of it” and additionally suggested adding keyword search to support more direct navigation. This perspective was also reflected in behavior; Block 3 logs showed task-driven tab selection with no single view dominating. Similarly, P2 explicitly requested additional analytical dimensions (e.g., “you can add more features, like a dataset, and then comparison of the accuracy in a table”), indicating a preference for comprehensive analytical coverage rather than interface minimalism. Rather than suggesting feature removal, participants framed their feedback around improving clarity and organization. P1 proposed integrating Surface and Results to reduce

perceived clutter. P4 recommended a progressive disclosure approach: ‘I would prefer to see fewer things and then be able to click and see more,’ highlighting a preference for more compact default views with optional expansion. These suggestions targeted presentation clarity rather than capability reduction; participants wanted the same functionality with lower initial cognitive overhead.

Theme 4: Trust and Provenance Needs. While participants generally found CiteView’s outputs accurate for the study tasks, several raised concerns about trusting AI-generated summaries in real research contexts. P11 articulated this most directly: “I don’t have a way of verifying whether the results that the tool is giving me is accurate.” P11 further noted that adding clearer source provenance “might be like the only improvement” needed. This represents a genuine limitation: even when outputs are accurate, perceived accuracy depends on transparency of sourcing. Long-term adoption would require more explicit provenance signals to support user verification beyond task performance.

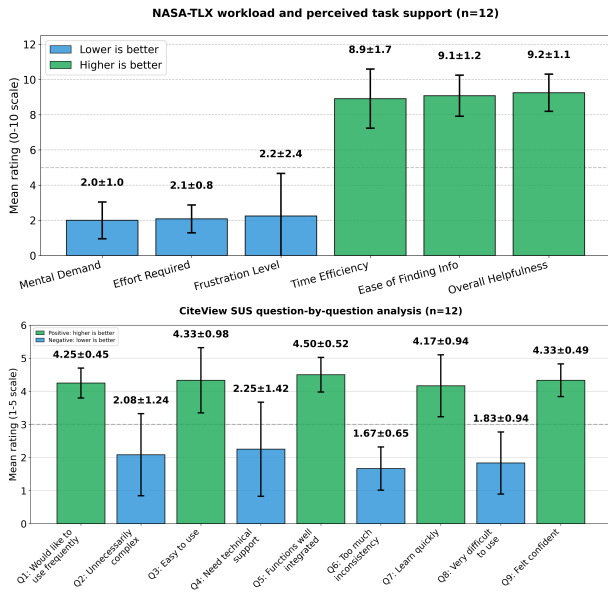


Figure 4: User evaluation results. (Top) NASA-TLX workload (lower is better) and supplementary domain-specific items (higher is better). (Bottom) SUS item-level ratings showing strong agreement on positive items and disagreement on negative items, consistent with overall usability ($M = 71.9$).

7 Discussion

Integrating citation analysis into reading enables effective sense-making without substantial cognitive burden. When viewed through information foraging theory [22], the preference for Results and Compare tabs reflect users seeking high-scent paths (concrete metrics and explicit differences) over lower-scent summaries. The sense-making model [26] explains why different tabs support different phases: Surface for orientation, Compare for schema refinement, Results for evidence marshaling. This alignment suggests the four-tab design reflects genuine cognitive structures, not incidental complexity. Consistent with this, the system achieved positive usability (SUS $M=71.9$) and low cognitive load despite substantial analytical depth. Suggestions focused on presentation (legends, naming, navigation) rather than functionality, with five participants stating that all features were “crucial.”

Contributions beyond the tool. Beyond the specific system, this work contributes design insights about how to support citation sensemaking in reading interfaces. First, our four-need framework (orientation, comparison, structural understanding, evidence verification) provides a reusable vocabulary for decomposing citation information needs, and future tools can use these categories to organize features even across different implementations. Second, the empirical finding that readers strongly prefer structured, insight-driven views over descriptive summaries challenges the common design assumption that AI reading assistants should primarily summarize. Our results suggest that *what* information is foregrounded (results, comparisons, visual evidence) matters more than *how much* information is provided, and that readers approach

citations with specific analytical goals rather than seeking comprehensive overviews. Third, the observed complementarity of the four views, each used for distinct question types in Block 3, provides empirical grounding for multi-view designs in scholarly tools. These findings generalize beyond CiteView and inform the design of future AI-augmented reading interfaces.

Baseline Rationale. The study does not include a formal baseline comparison. No equivalent system supports in-situ citation analysis within a reading interface; tools such as Google Scholar and standard PDF readers address discovery and access, not contextual explanation or structured comparison. A task-matched baseline would also have doubled session length, conflating interface design with retrieval latency and access barriers. Following established practice for novel interaction paradigms in HCI [13, 15, 28], we report viability evidence rather than superiority claims.

Design Implications. (1) *Prioritize insight-driven views:* foreground results, comparisons, and evidence over summaries. P1: “results would be the most favourite because I can instantly understand what the cited paper did.” Future interfaces might organize by analytical purpose (“Verify Claims,” “Compare Methods”) rather than content type. (2) *Progressive disclosure:* surface core insights first with clear pathways to depth, as P5 noted the risk of “getting lost.” (3) *Question-driven navigation:* support both structured exploration and in-tool search, as P6 suggested: “If I could just search within CiteView... it gives you an option to go to figures.”

8 Limitations and Future Work

Trust, Provenance, and Hallucination Risk. CiteView mitigates hallucination through evidence-grounded retrieval, structured output constraints, and rule-based fallbacks. Our pipeline validation found no factual inaccuracies across 60 surface bullets or 90 compare dimension pairs. Nevertheless, participants identified trust as a key barrier to adoption beyond controlled settings. P11 articulated this directly: “I don’t have a way of verifying whether the results that the tool is giving me are accurate,” adding that clearer source provenance “might be like the only improvement” needed. This points to an important distinction: even when outputs are factually correct, perceived accuracy depends on transparency of sourcing. Future work should therefore add interactive provenance, allowing users to click any AI-generated claim to reveal the underlying passage from the cited paper, bridging the gap between output accuracy and user confidence. **Full-Text Dependence.** CiteView’s analysis depth is contingent on open-access availability. When full text cannot be retrieved, the system degrades to metadata-only analysis and notifies the reader. Future work could develop metadata-enriched surrogates, using abstracts, citation contexts, and structured metadata, to maintain analytical coherence when full text is unavailable. **Long-Term Adoption.** Field deployments are needed to assess integration into sustained workflows and whether multi-modal interfaces encourage shallow engagement that bypasses deep reading. During development, outputs were manually verified by two researchers across 15 citation analyses (see Section 4). However, this does not constitute a formal evaluation; systematic assessment of extraction precision, bullet-point correctness, and graph fidelity against ground truth remains future work, particularly for semantic graph generation.

9 Conclusion

We introduced CiteView, a system integrating multi-modal citation analysis within PDF reading. Four views (Surface, Compare, Viz, Results) grounded in sensemaking theory enable readers to understand cited papers without leaving their context. A RAG pipeline combining deterministic parsing, relevance scoring, and multi-agent LLM synthesis grounds outputs in retrieved evidence. Our study (n=12) demonstrated high accuracy (97.9%), low cognitive load, and a clear preference for insight-driven views. Citation tools should prioritize contribution-level analysis, employ progressive disclosure, and support both structured and question-driven exploration.

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